

Does a Wider Trail Fix the 1x ATR Whipsaw?

Single-variant test: partial lock + 1.5x ATR trail, vs. Phase 3B's 1x ATR and fixed-exit baseline — same 6 fixed windows, floor 60

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Phase 3B found Condition C (partial 50% lock at 1R + 1x ATR trailing remainder) directionally confirmed that Window 5's problem is exit management, but at a net cost elsewhere, leaving the combined 15m+1h projection slightly below the fixed-exit baseline (+22.5% vs +23.6%). The identified failure mechanism: 1x ATR is tight enough on 15m's noise that the trail gets hit almost immediately after the stop reaches breakeven, before a trade has room to develop. This batch tests the direct fix — widen the trail to 1.5x ATR, keeping the partial lock (proven load-bearing in Phase 3B) — as a single targeted variant, not a new exploration.

Architecture Check

'atr1.5' already existed as a valid `TrailingStopMode` value (type `TrailingStopMode` = 'fixed' | 'atr1' | 'atr1.5') and was already wired into the trail-multiplier calculation. **No code changes were needed for this batch** — purely configuration, same as Phase 3B's Condition C.

Three-Condition Comparison

CONDITION	DESCRIPTION	MEAN RETURN	MEAN PF	MEAN WR	PROFITABLE SUB-RUNS	MEAN MAX DD	MEAN HOLD TIME
A	Floor 60, fixed exit (baseline)	+20.4%	1.12	51.0%	10/18 (56%)	19.5%	n/a*
C	Partial 50% + 1x ATR trail	+18.2%	1.11	51.5%	11/18 (61%)	19.0%	7.91
D (new)	Partial 50% + 1.5x ATR trail	+18.5%	1.11	51.4%	10/18 (56%)	20.0%	8.31

* Condition A predates hold-time tracking (added in Phase 3B); not re-run to backfill, consistent with the discipline applied throughout this project.

D edges out C by a trivial 0.30 percentage points (18.45% vs 18.15%) and remains below A (20.42%). Profit factor is identical across all three to two decimal places (1.11-1.12) — this is not a dramatic shift in either direction on 15m alone.

Per-Window Return

WINDOW	REGIME	A (FIXED EXIT)	C (1X ATR)	D (1.5X ATR)
1 (2020-10→2021-03)	Bull	+21.5%	+19.8%	+20.2%
2 (2023-10→2024-03)	Bull	+27.3%	+21.9%	+26.2%
3 (2021-05→2021-10)	Bear	+22.1%	+18.0%	+20.3%
4 (2022-04→2022-09)	Bear	+31.3%	+25.2%	+27.3%
5 (2023-07→2023-12)	Choppy	-14.9%	-12.0%	-12.7%
6 (2024-04→2024-09)	Choppy (sub.)	+35.2%	+36.0%	+29.4%

D recovers most of C's losses on Windows 2-4 (closer to A again) but gives back ground on Window 6 specifically (+29.4% vs C's +36.0%, A's +35.2%) — the wider trail let a strong trending move in Window 6 get trailed out earlier than either the fixed target or the tighter 1x trail did. Window 5 itself is essentially unchanged from C (-12.7% vs -12.0%), within noise of each other.

Window 5 Isolated

CONDITION	WINDOW 5 RETURN	WINDOW 5 PF	WINDOW 5 WR
A	-14.9%	0.87	42.8%
C	-12.0%	0.88	44.8%
D	-12.7%	0.87	44.4%

Proportional Movement — Does D Still Help Window 5 Disproportionately?

CONDITION	WINDOW 5 Δ VS A (PP)	WINDOWS 1-4+6 AVG Δ VS A (PP)
D	+2.2	-2.8

The same directional pattern from Condition C persists at 1.5x ATR: Window 5 still moves positive relative to the fixed-exit baseline (+2.2pp) while the other five windows move slightly negative on average (-2.8pp). The diagnostic mechanism is real and reproduces at a different trailing distance — this is not an artifact of the specific 1x ATR value tested in Phase 3B.

Hold Time — Did the Wider Trail Actually Change Exit Timing?

CONDITION	MEAN HOLD TIME (CANDLES)
B (Phase 3B reference — no partial, 1x ATR)	7.89
C (partial + 1x ATR)	7.91
D (partial + 1.5x ATR)	8.31

This is the clearest confirmation in this report that the lever is doing what it's supposed to. B and C had nearly identical hold times (7.89 vs 7.91) despite wildly different outcomes — the trail distance wasn't the variable separating them, the partial lock was. D's hold time (8.31) is meaningfully longer than both — about 5% more candles held on average — confirming the wider trail genuinely gives trades more room before exiting, exactly the mechanism the brief predicted. The whipsaw is measurably reduced. It just isn't enough to flip the aggregate result in D's favor.

Ceiling Check

CONDITION	BEST INDIVIDUAL SUB-RUN
A	+108.0% (SOLUSDT, Window 6)
C	+108.0% (SOLUSDT, Window 6)
D	+106.5% (SOLUSDT, Window 6)

D's ceiling is essentially preserved — a 1.5-point difference on a 108-point outcome is within the kind of run-to-run variation expected from a single sub-run, not a material drop. The wider trail did not meaningfully cap the best case, consistent with (though not dramatically better than) C.

Combined 15m + 1h Projected Impact

CONFIGURATION	COMBINED MEAN RETURN	WINDOWS ≥30%
A — Floor 60 + 1h (current best, fixed exit)	+23.6%	2/6
C — Partial + 1x ATR + 1h	+22.5%	2/6
D — Partial + 1.5x ATR + 1h	+22.7%	2/6

This is the number that arbitrates the verdict, not the 15m-only direction. **D (+22.7%) improves marginally on C (+22.5%) but does not surpass A (+23.6%).**

Explicit Verdict

OUTCOME 2 applies: D beats Condition C but not the fixed-exit baseline A. 1.5x ATR improved on 1x ATR — hold time confirms the whipsaw mechanism is real and is measurably reduced (8.31 vs 7.91 candles, ~5% longer), and the diagnostic pattern (Window 5 helped disproportionately relative to other windows) reproduces cleanly at

this distance too. But the net combined effect (+22.7%) still falls short of simply keeping a fixed target (+23.6%).
The whipsaw was reduced but not enough to net-improve over the existing exit logic.

The exit-logic family of levers has now been exhausted at the 1x and 1.5x ATR distances for the partial-lock configuration. Three exit-logic configurations have been tested against the same 6 windows (no lock + 1x trail, partial lock + 1x trail, partial lock + 1.5x trail) without any of them beating the fixed-exit baseline. Per this batch's own scope, distances beyond 1.5x ATR are not being tested next — that would be diminishing-returns tuning on a lever family that has not been productive. **Recommendation: keep the fixed-exit floor-60 configuration as the working 15m baseline, and pivot to choppy-regime-conditional position sizing as the next, structurally different lever** — smaller position size specifically when the regime classifier detects ranging conditions, full size in trending regimes. This accepts that choppy regimes will underperform on a relative basis and manages the downside through sizing rather than continuing to search for an exit-mechanics fix that three tested variants have not delivered.

Appendix — All 54 Sub-Run Results (Conditions A, C, D)

WINDOW	SYMBOL	CONDITION	TRADES	WR	PF	RETURN	MAX DD	AVG HOLD (CANDLES)
1	BTCUSDT	A	164	52.4%	1.15	+18.5%	16.6%	n/a*
1	ETHUSDT	A	177	55.4%	1.07	+8.0%	12.2%	n/a*
1	SOLUSDT	A	137	56.2%	1.31	+38.0%	12.3%	n/a*
1	BTCUSDT	C	171	51.5%	1.05	+6.0%	20.2%	7.5
1	ETHUSDT	C	179	55.9%	1.11	+14.0%	10.9%	7.5
1	SOLUSDT	C	136	55.9%	1.34	+39.5%	12.4%	5.7
1	BTCUSDT	D	169	52.7%	1.13	+17.1%	18.6%	7.9
1	ETHUSDT	D	178	55.6%	1.05	+6.4%	11.6%	7.8
1	SOLUSDT	D	136	55.9%	1.32	+37.2%	12.4%	6.1
2	BTCUSDT	A	191	46.6%	0.77	-23.7%	24.6%	n/a*
2	ETHUSDT	A	171	49.1%	0.98	-2.0%	23.6%	n/a*
2	SOLUSDT	A	186	57.0%	1.68	+107.6%	15.6%	n/a*
2	BTCUSDT	C	190	46.8%	0.82	-21.8%	23.0%	9.1
2	ETHUSDT	C	181	50.8%	1.02	+2.9%	23.4%	8.0
2	SOLUSDT	C	186	57.0%	1.52	+84.6%	11.0%	7.2
2	BTCUSDT	D	191	46.6%	0.77	-23.8%	25.1%	9.3
2	ETHUSDT	D	172	50.0%	0.98	-2.8%	23.9%	8.5
2	SOLUSDT	D	189	57.7%	1.65	+105.2%	12.3%	7.7
3	BTCUSDT	A	181	47.0%	0.82	-20.7%	28.4%	n/a*
3	ETHUSDT	A	153	56.2%	1.34	+50.8%	24.6%	n/a*
3	SOLUSDT	A	149	61.7%	1.33	+36.2%	14.1%	n/a*
3	BTCUSDT	C	184	46.7%	0.81	-22.4%	30.7%	8.5
3	ETHUSDT	C	157	56.7%	1.28	+40.9%	26.9%	8.4
3	SOLUSDT	C	150	61.3%	1.32	+35.5%	15.0%	7.1

3	BTCUSDT	D	178	47.2%	0.80	-22.5%	30.5%	9.0
3	ETHUSDT	D	156	57.7%	1.35	+51.2%	26.8%	8.8
3	SOLUSDT	D	150	61.3%	1.29	+32.1%	15.0%	7.5
4	BTCUSDT	A	177	45.8%	0.95	-6.7%	23.6%	n/a*
4	ETHUSDT	A	186	54.8%	1.39	+55.6%	17.5%	n/a*
4	SOLUSDT	A	201	53.2%	1.23	+45.1%	14.9%	n/a*
4	BTCUSDT	C	181	44.8%	0.86	-17.1%	24.8%	8.2
4	ETHUSDT	C	193	54.4%	1.31	+43.2%	14.8%	7.7
4	SOLUSDT	C	189	55.0%	1.28	+49.4%	14.9%	6.6
4	BTCUSDT	D	184	44.0%	0.86	-17.1%	25.2%	8.7
4	ETHUSDT	D	190	54.7%	1.35	+50.9%	15.9%	8.2
4	SOLUSDT	D	189	54.5%	1.28	+48.2%	13.6%	6.7
5	BTCUSDT	A	172	43.0%	0.93	-9.1%	23.3%	n/a*
5	ETHUSDT	A	150	41.3%	0.79	-21.5%	24.7%	n/a*
5	SOLUSDT	A	188	44.1%	0.89	-14.2%	23.3%	n/a*
5	BTCUSDT	C	175	42.9%	0.92	-10.9%	20.9%	8.7
5	ETHUSDT	C	154	42.9%	0.74	-23.4%	27.3%	8.4
5	SOLUSDT	C	193	48.7%	0.99	-1.8%	18.2%	8.1
5	BTCUSDT	D	180	42.8%	0.93	-10.3%	27.1%	9.2
5	ETHUSDT	D	152	42.1%	0.72	-24.0%	27.6%	8.8
5	SOLUSDT	D	190	48.4%	0.97	-3.8%	19.0%	8.4
6	BTCUSDT	A	174	54.6%	1.21	+26.7%	12.9%	n/a*
6	ETHUSDT	A	170	41.8%	0.69	-29.0%	29.5%	n/a*
6	SOLUSDT	A	194	58.0%	1.66	+108.0%	10.0%	n/a*
6	BTCUSDT	C	180	53.3%	1.18	+22.8%	14.9%	8.1
6	ETHUSDT	C	181	44.8%	0.82	-22.7%	23.9%	8.6
6	SOLUSDT	C	200	56.8%	1.66	+108.0%	8.7%	9.2
6	BTCUSDT	D	185	53.0%	1.07	+9.0%	17.4%	8.2
6	ETHUSDT	D	174	43.1%	0.74	-27.4%	28.0%	9.1
6	SOLUSDT	D	203	57.4%	1.62	+106.5%	10.4%	9.5

* Condition A predates hold-time tracking.

Report generated from 18 newly-run live simulations (Condition D) plus existing saved data for Conditions A (Phase 3 Score Floor Test) and C (Phase 3B Exit Logic Test), against real historical Binance market data. Zero console errors. Raw data: `training-exports/phase3c-conditionD-results.json` , `training-exports/phase3b-conditionC-results.json` , `training-exports/phase3-floor60-results.json` .